

01 · Who should we email? — customer-level uplift [ANCHOR]

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The decision. We have a €10 discount and a customer list. Emailing everyone burns margin on people who'd have bought anyway (and nudges some to unsubscribe); emailing no one leaves money on the table. **Who, exactly, should get the offer?**

Every customer is one of four types, and only the treatment *effect* — not the outcome — tells them apart:

	buys if emailed	doesn't buy if emailed
buys if not emailed	(yellow) <i>Sure Thing</i> ($\tau \sim 0$, wasted discount)	(red) <i>Sleeping Dog</i> ($\tau < 0$, email backfires)
doesn't buy if not emailed	(green) <i>Persuadable</i> ($\tau > 0$, the target)	(white) <i>Lost Cause</i> ($\tau \sim 0$, no point)

Targeting is worth doing **only because the effect is heterogeneous** — if everyone had the same τ you'd rationally treat all or none. So the entire problem is estimating the per-customer effect $\tau(x)$ well, and turning it into a euro decision under uncertainty.

This is the flagship notebook: the fixed **7-step contract**, then the **three depths** a data scientist needs to defend the number.

- **Steps** 1 question -> 2 simulate -> 3 identify -> 4 estimate -> 5 validate -> 6 decide in € -> 7 caveats
- **Depth A** estimator bake-off & failure modes · **Depth B** identification rigour & sensitivity · **Depth C** euro policy, VOI & test design

Narration is business-first. “Bayesian” stays in the background; credible intervals show up as “*how sure are we, and does it change what we'd do?*”

FAST=True N=900 sampling={'draws': 140, 'tune': 140, 'chains': 2, 'm': 30}

1.1 2 · Simulate a ground truth (validation-first)

We can't check a causal method on real data — the individual effect is never observed. So we make the method **prove itself on data where we planted the answer**, then (with the same code) swap in real data. Our simulator plants a known per-customer effect $\tau(x)$, a known baseline

spend, and — crucially for the observational case — a **selection rule**: historically marketers emailed the already-engaged, recently-active customers, so treatment is *confounded* with the kind of customer.

The planted effect is deliberately realistic and **heterogeneous**: engaged mid-recency customers are **persuadable** (large $+\tau$); habitual high-frequency low-engagement customers are **sleeping dogs** ($\tau < 0$, the email pushes them to unsubscribe); the rest sit near zero.

Swap in real data here. Replace this cell with a loader returning the same columns; `cmp.data.load_hillstrom` gives a real *randomized* email experiment.

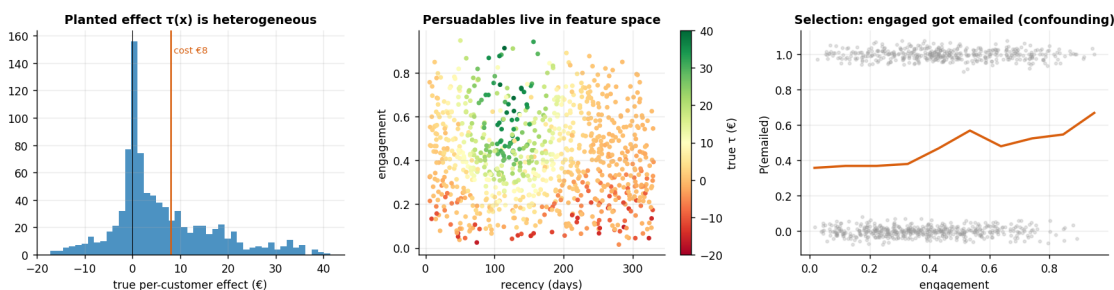
True ATE = €6.07 · emailed historically: 47% of base

Uplift types (share of base):

sure thing / lost cause	0.39
persuadable	0.32
sleeping dog	0.29

	recency	frequency	monetary	tenure	engage	T	y \
0	208.156027	6.0	108.310843	13.737214	0.276899	0.0	42.175390
1	296.594485	3.0	142.587867	25.200485	0.473980	0.0	67.436225
2	257.097849	4.0	107.168786	57.141597	0.390539	0.0	48.743181
3	78.192337	7.0	98.732666	13.421173	0.086593	0.0	46.782878
4	102.554043	7.0	87.199065	4.999605	0.359222	1.0	69.098355

	tau
0	-0.457966
1	-0.028135
2	0.129894
3	-9.869026
4	21.912770



1.2 3 · Identify — estimand and assumptions (kept separate from estimation)

Estimand ladder (three questions, three objects):

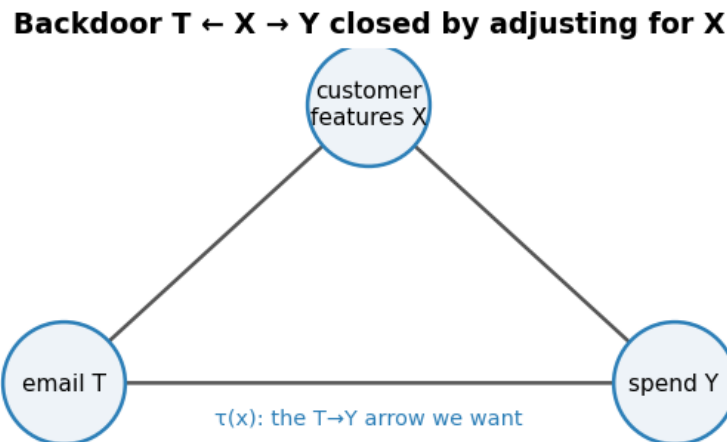
object	definition	used for
ATE	$\mathbb{E}[Y(1) - Y(0)]$	“does the campaign work at all?”
CATE $\tau(x)$	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$	<i>who</i> to target
policy value	$\mathbb{E}[(Y(1) - Y(0) - c) \mathbb{1}_{\{\text{target } x\}}]$	euros from a targeting rule

The uplift decision needs the whole ladder, but everything rests on $\tau(x) = \mu_1(x) - \mu_0(x)$.

Identification assumptions (turn the estimand into something estimable):

1. **Unconfoundedness** $\{Y(1), Y(0)\} \perp T \mid X$ — given the features, who got emailed is as-good-as-random. *Holds by design under an A/B test; an assumption otherwise.* (Depth B)
2. **Positivity / overlap** $0 < e(x) < 1$, $e(x) = P(T=1 \mid X=x)$. Checked before we trust any adjustment. (Depth B)
3. **SUTVA** — no spillover between customers, one version of the offer.

Under 1–3 the g-formula gives $\tau(x) = \mu_1(x) - \mu_0(x)$. **Identification is an assumption about the world (untestable); estimation is where the model earns its keep — priors never rescue a broken identification strategy.** The adjustment set is the DAG’s backdoor set:



1.3 4 · Estimate — overlap first, then a posterior over each effect

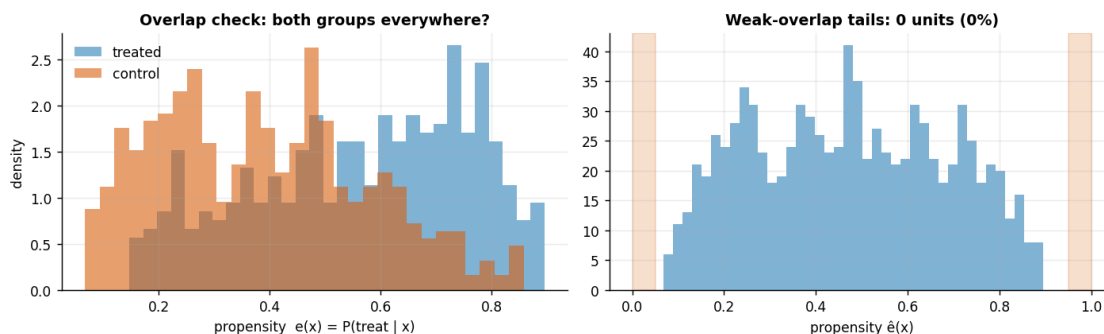
Discipline: **check overlap before estimating**, because adjustment on regions with no control (or no treated) is extrapolation dressed up as inference. Then we fit two ways and cross-check:

- **BCF** (Bayesian Causal Forest) — flexible BART response surface with the estimated propensity fed into the *prognostic* part to absorb targeted-selection bias, and a tighter prior on the *treatment* part so heterogeneity shrinks toward homogeneity rather than being read out of noise. Our primary CATE estimator here.

- **AIPW** (augmented IPW) — a model-agnostic **doubly-robust** ATE: consistent if *either* the outcome model or the propensity model is right. A sanity check on the ATE that leans on completely different machinery than BART.

Baked-in gotcha. Scoring BART counterfactuals silently returns *frozen* training predictions — every effect exactly 0 — unless the tree node is resampled (`sample_vars=["mu"]`). `cmp.estimators` always does this; a test asserts non-zero CATE.

overlap adequate - 0 of 900 units in the <0.05 / >0.95 tails



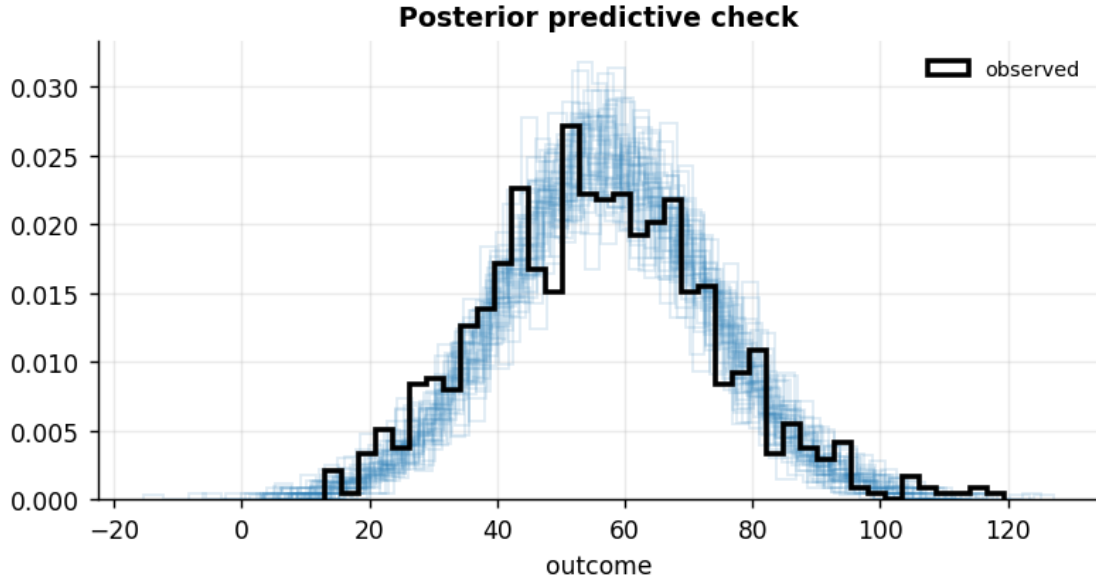
BCF ATE = €6.20 (true €6.07)

AIPW ATE = €5.85 90% CI [5.19, 6.99] (doubly-robust, 0 trimmed)

Two independent estimators agreeing on the ATE is our first credibility check.

Model criticism — posterior predictive check. Before reading effects off the model, ask whether it can even reproduce the data it was fit on. We overlay observed spend against posterior-predictive replicates; systematic misfit (wrong spread, missed skew) would warn us the effect estimates aren't trustworthy.

Observed spend sits within the replicate envelope (residual sigma ~ €15) - no gross misfit.

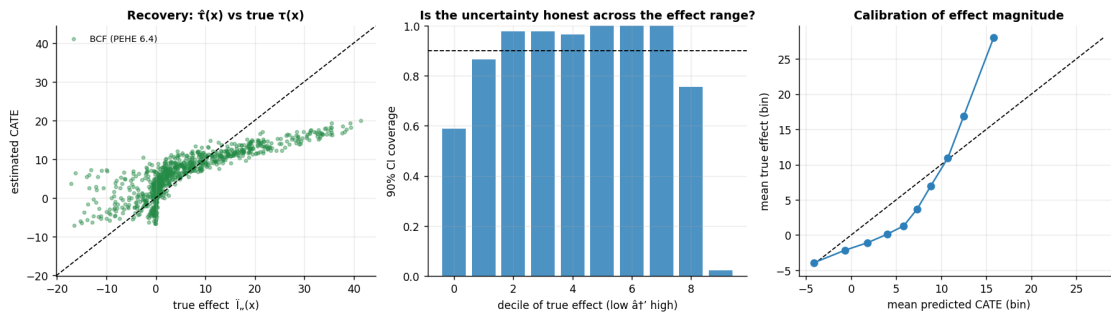


1.4 5 · Validate — recover truth, honest uncertainty, and ranking quality

Because we planted the truth we can run the full validation battery. Three questions:

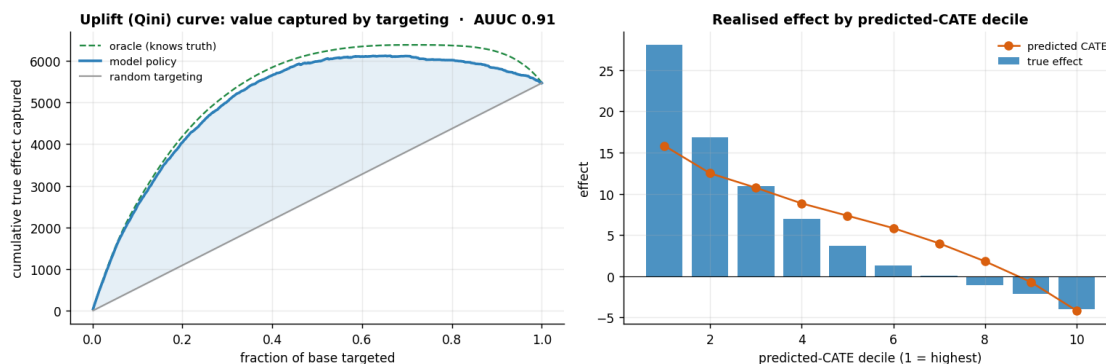
1. **Recovery & calibration** — does $\tau_{\text{hat}}(x)$ track $\tau(x)$ (PEHE, correlation)? Do the credible intervals contain the truth $\sim 90\%$ of the time, *across the whole effect range* (coverage by decile), and are the *magnitudes* calibrated (reliability curve)? And are the intervals **sharp** enough to act on?
2. **Ranking quality** — for targeting, ordering customers correctly matters more than exact magnitudes. The **Qini/uplift curve** and **AUUC** measure exactly that, vs a random and an oracle policy. The **uplift-by-decile** staircase shows the top deciles really do have the biggest true effect.
3. **Stability** — is the recovery a fluke of one seed? We refit on several resimulations.

PEHE 6.37 · corr 0.84 · 90% coverage 82% · interval sharpness €14.6



AUUC = 0.91 of oracle (1.0 = perfect ranking, 0 = no better than random).

Top decile true effect €28.1 vs bottom €-3.9 - the model concentrates real uplift where we'd spend.



seed	ATE_hat	true_ATE	PEHE	AUUC
100	6.78	5.74	6.64	0.88
101	5.13	5.83	6.42	0.89
102	5.88	5.44	5.93	0.89
103	6.17	5.20	6.64	0.88

ATE recovery bias across seeds: +0.44 (+/-0.81) - stable, not a lucky seed.

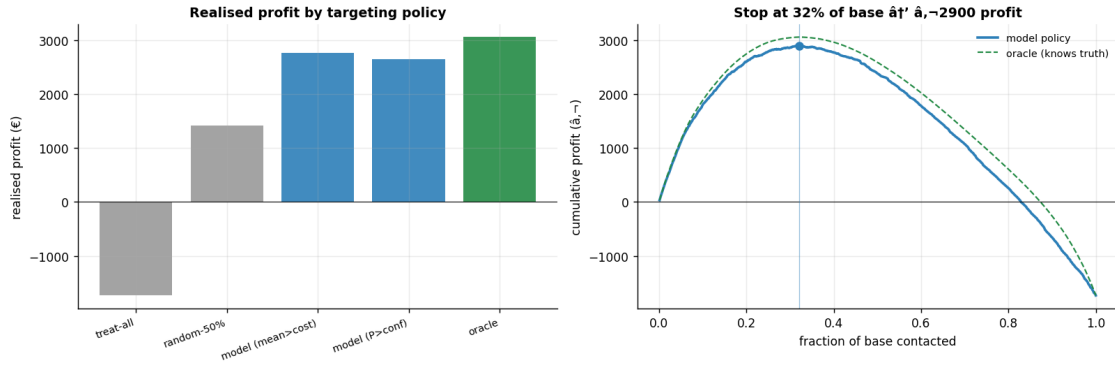
1.5 6 · Decide, in euros

The posterior only matters once it becomes a number a manager acts on. Three moves:

- **Compare policies on realised profit.** treat-all, random, “mean effect beats cost”, the honest confidence rule, and the oracle — evaluated on the known truth. This is the slide that shows targeting *beats treat-all in euros* and how close we get to the oracle.
- **Whom to target — the honest rule.** Not “mean beats cost” but $P(\tau(x) > \text{cost})$ high. A customer whose point estimate says target but whose interval spans the cost line is a coin-flip, not a yes.
- **Profit curve & optimal stop** — rank by effect, contact down the list, stop at the peak.

	policy	profit	frac_contacted	profit_vs_all
	treat-all	-1734.0	1.0	0.0
	random-50%	1414.0	1.0	3148.0
model	(mean>cost)	2770.0	0.0	4505.0
	model (P>conf)	2651.0	0.0	4385.0
	oracle	3058.0	0.0	4792.0

Emailing everyone: €-1,734 · honest model rule: €2,651 (87% of oracle) · targeting 184 of 900 customers.



1.6 7 · Caveats — the honest failure modes

- **Unconfoundedness is untestable.** If a *hidden* driver moved both who got emailed and how much they spend, the estimate drifts. Depth B quantifies exactly how much it would take to flip the call, as an E-value.
- **Overlap matters.** If some kind of customer was *always* emailed, we extrapolate rather than compare. We checked the tails above and can trim.
- **Estimator choice is not cosmetic.** The S-learner regularises the treatment signal away and under-detects heterogeneity — Depth A shows it failing on PEHE *and* AUUC.
- **BCF ~ T-learner when confounders are fully observed.** The propensity trick earns its keep specifically under targeted selection; don't oversell it.
- **Sleeping dogs are real.** Negative-effect customers mean “treat everyone” can be actively value-destroying, not merely wasteful — which is exactly what the policy comparison showed.

2 Depth A · Estimator bake-off & failure modes

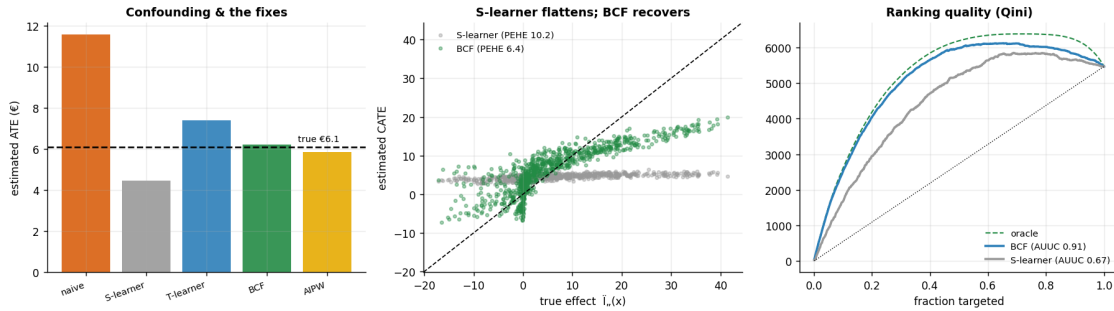
Three estimators x two regimes, judged against the known truth on the metrics that matter for targeting — **PEHE** (magnitude error), **AUUC** (ranking quality), **ATE bias**, and **coverage** (honest uncertainty):

- **S-learner** — one surface $f(x, t)$; regularises the treatment term -> **flattens heterogeneity** (the labelled failure mode).
- **T-learner** — separate surfaces; solid default under randomization.
- **BCF** — propensity-aware, shrink-to-homogeneity; best under confounding.

In the **randomized** regime unconfoundedness holds by design; in the **observational** regime, targeted selection confounds treatment, and the naive “emailed — not-emailed” average is badly biased. This is the slide that earns the method its keep.

Naive 'emailed - not emailed' (observational): €11.6 vs true €6.1 -> bias +5.5
(confounding nearly doubles it)

	regime	estimator	PEHE	corr	AUUC	ATE_bias	cov90
0	randomized	S-learner	10.04	0.60	0.67	-2.07	0.76
1	randomized	T-learner	6.22	0.83	0.89	0.05	0.86
2	randomized	BCF	7.28	0.77	0.86	-0.06	0.80
3	observational	S-learner	10.20	0.60	0.67	-1.62	0.75
4	observational	T-learner	5.72	0.86	0.92	1.32	0.93
5	observational	BCF	6.37	0.84	0.91	0.13	0.82



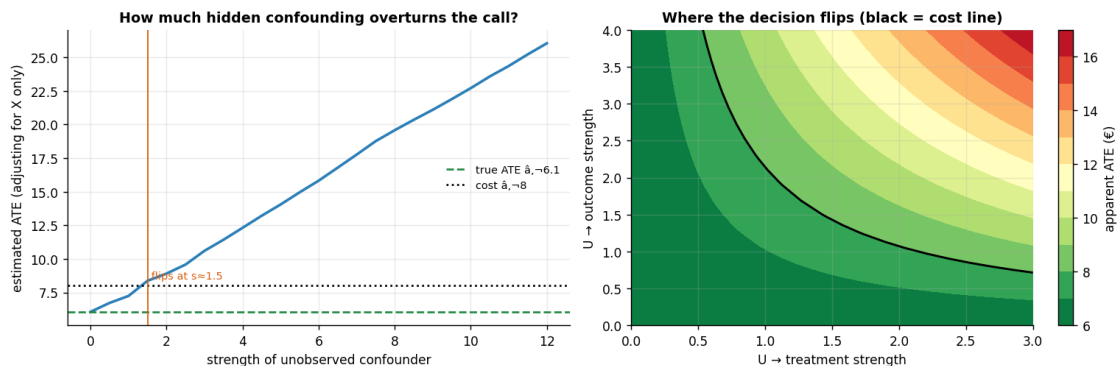
Read-out. Randomized: all three are ~unbiased on the ATE, but the S-learner already ranks worse (lower AUUC, higher PEHE). Observational: the naive average is badly biased; the S-learner both misses heterogeneity *and* is mis-calibrated; **BCF gets the closest per-customer recovery, the best ranking, and near-nominal coverage** — and it agrees with the completely separate AIPW estimator on the ATE. Right ranking + honest uncertainty is what the euro decision rests on.

3 Depth B · Identification rigour & sensitivity

Adjustment only works where treated and control coexist, and only if we measured the confounders that matter. We already checked **overlap** (step 4). The deeper question: unconfoundedness is *untestable*, so **how fragile is the conclusion to something we didn't measure?**

We introduce a hidden driver U that moves *both* who gets emailed and how much they spend, sweep its strength on **two** axes ($U \rightarrow$ treatment and $U \rightarrow$ outcome), refit adjusting for observed **X only** (as we would in reality, blind to U), and map where the decision flips. Then we summarise the fragility as a single **E-value**: the minimum confounder strength that would be needed to explain the effect away below the cost line.

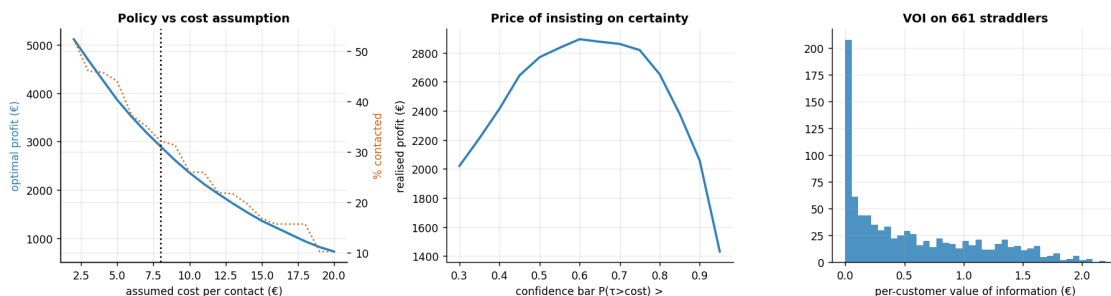
Tipping point (1-D): confounder strength ~ 1.5 . E-value ~ 5.76 - an unmeasured confounder would need $\sim 5.8x$ association with BOTH email and spend (on the risk-ratio scale) to explain the effect down to break-even. Ask the domain expert whether anything that strong is plausible; if not, the result is robust.



4 Depth C · Euro policy — sweeps, value of information & test design

The decision is a number *under assumptions*. We stress the two managers argue about — the per-contact **cost** and how **confident** we insist on being — then price the remaining uncertainty and turn it into a concrete **experiment design**.

Total VOI €508; €502 of it sits on the 661 'straddlers' (73% of base) whose interval spans the cost line.



4.0.1 The one-paragraph decision

Targeting the model's confident persuadables makes materially more money than emailing everyone (which here *loses* money to sleeping dogs), and captures most of the oracle's profit. The call is robust to plausible cost assumptions and would only reverse under fairly strong hidden confounding (Depth B's E-value). The remaining uncertainty is worth real money and is concentrated on the "straddler" customers whose interval crosses the cost line — **so the follow-up is a small randomized A/B test on the straddlers, not a blanket send**. Act where we're sure; measure where we're not. (*FAST mode uses a coarse posterior, so the straddler set is wide; a full run tightens the intervals and shrinks it.*)

```
{  
  "true_ate": 6.0732099315101244,  
  "bcf_ate": 6.201872997920628,  
  "aipw_ate": 5.851767331319051,  
  "PEHE": 6.368359564219455,  
  "AUUC": 0.9140155140227195,  
  "coverage90": 0.8155555555555556,  
  "profit_treat_all": -1734.1110616408864,  
  "profit_model": 2650.865750738045,  
  "profit_oracle": 3057.9390084417983,  
  "sensitivity_tip": 1.5,  
  "e_value": 5.756405386243919,  
  "VOI_total": 507.8040944716226,  
  "n_straddlers_to_test": 661  
}
```